



Chapter 4

Physical Transformations of Pure Substances



Thermodynamics  **Level 1** The phase transition of pure substances


Without change of composition

Phase diagrams

A map of the pressures and temperatures at which each phase of a substance is the most stable.

- I. Interpretation
- II. Determine boundaries
- III. Liquid surface



4.1 Phase

1. Phase

A **phase** of a substance is a form of matter that is uniform throughout in **chemical composition** and **physical state**.

Gas Liquid Solid



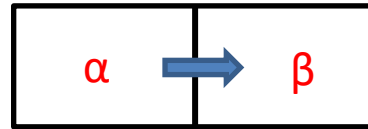
2. Phase transition

A phase transition is the spontaneous¹ conversion of one phase into another phase², occurs at the specific temperature for a given pressure³.

characteristics¹⁻³

Chemical potential (μ)

The potential that a substance has for bringing about change in a system.



Closed system }
Open system } Matter exchange

$f(T, P)$

μ (chemical potential)

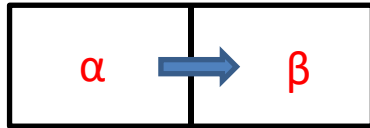
$\mu = \left(\frac{\partial G}{\partial n_i} \right)_{T, P}$ pure substance

$\mu = G_m$

$f(T, P, n_b, n_c, \dots)$

$\mu = \left(\frac{\partial G}{\partial n_i} \right)_{T, P, n_A, n_B, \dots}$ mixture

$\mu \neq G_m$



$$T, P, w' = 0$$

$$\Delta G = \sum \mu_i dn_i$$

$$\Delta G = \mu_\alpha dn_\alpha + \mu_\beta dn_\beta$$

$$-dn_\alpha = dn_\beta$$

$$\Delta G = (\mu_\alpha - \mu_\beta) dn_\alpha < 0$$

$$dn_\alpha < 0$$

$$\mu_\alpha - \mu_\beta > 0$$

$$\mu_\alpha \geq \mu_\beta \quad \begin{array}{l} > \text{spontaneous} \\ = \text{equilibrium} \end{array}$$



3. T_{trs}

A transition temperature, T_{trs} , represents the temperature at which the two chemical potentials are equal and the two phases are in equilibrium at the specific pressure.



4. Metastable phases

Thermodynamically unstable phases that persist because of the transition is kinetically hindered.

At normal T and P

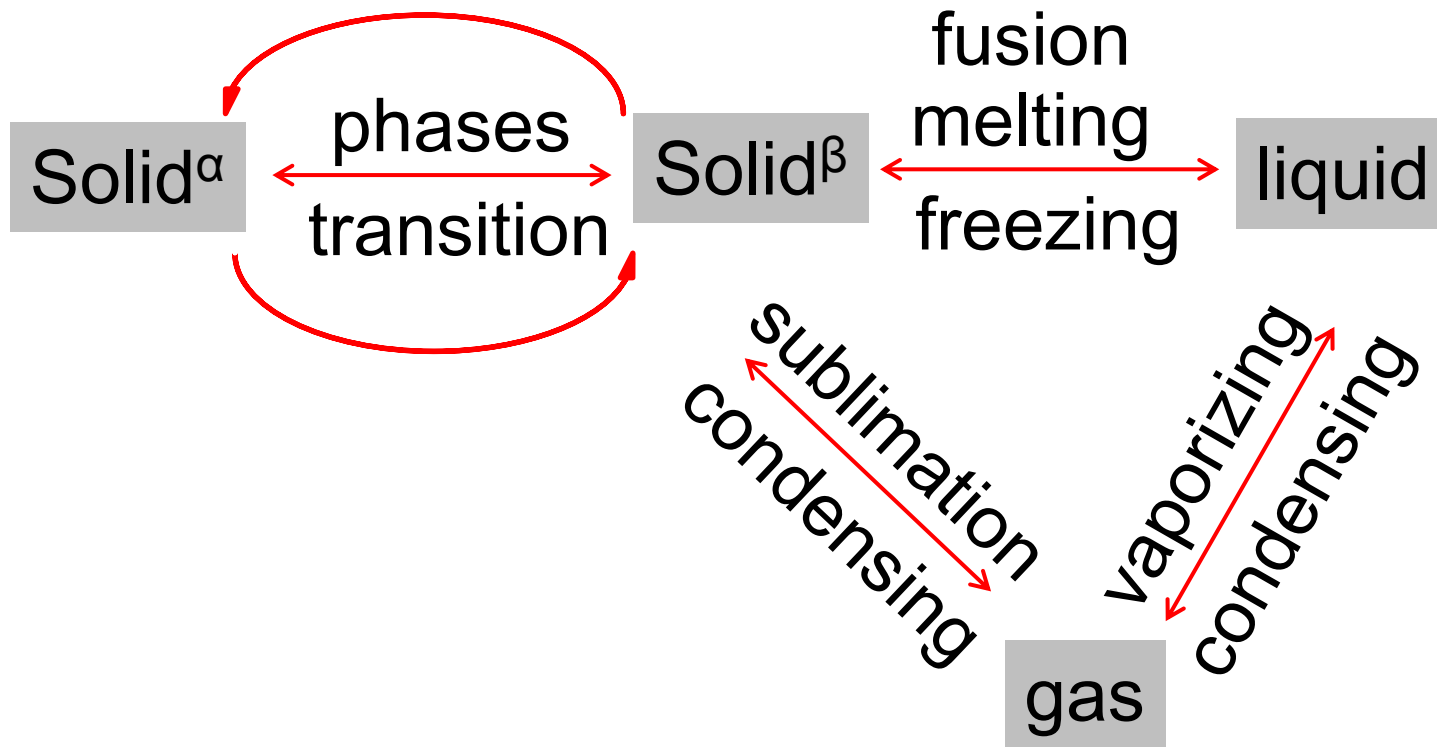
$$\mu(\text{diamond}) > \mu(\text{graphite})$$

Diamond is a metastable phase.



4.2 Phase boundaries

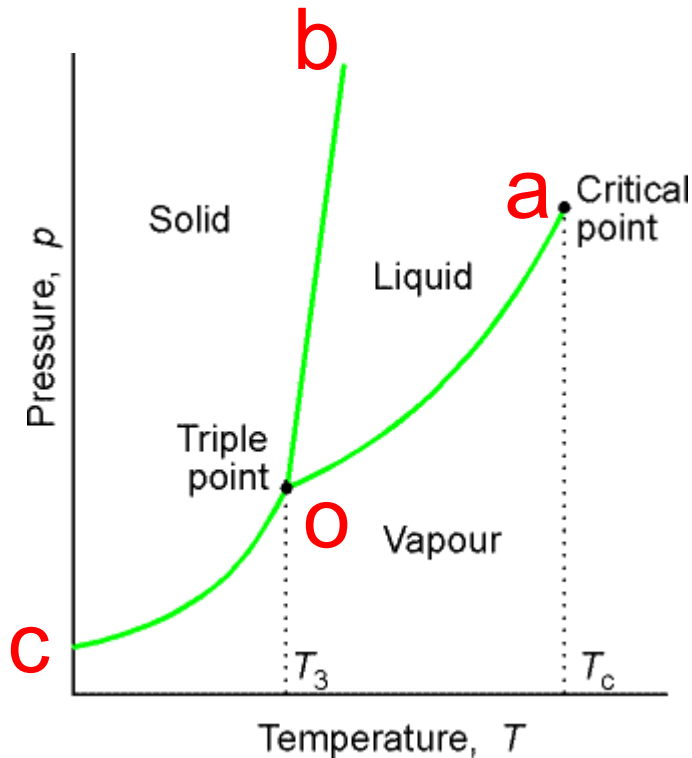
Phase transition:





1. Phase boundaries

The lines show the values of p and T at which two phases coexist in equilibrium.



The line **O-a** boiling points
The liquid-vapour phase boundary

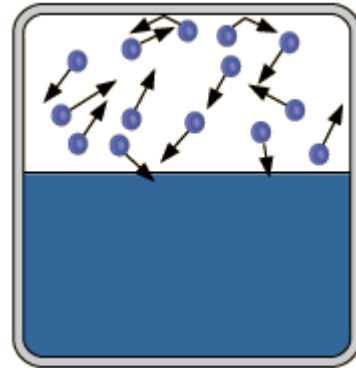
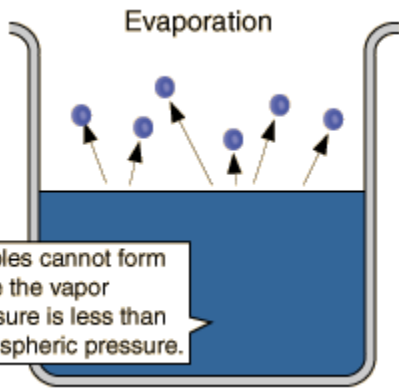
The line **O-b** melting points
The liquid-solid phase boundary

The line **O-c** sublimation points
The solid-vapour phase boundary

The point **O**: triple point

The point **a**: critical point

2. Vapour pressure and boiling points



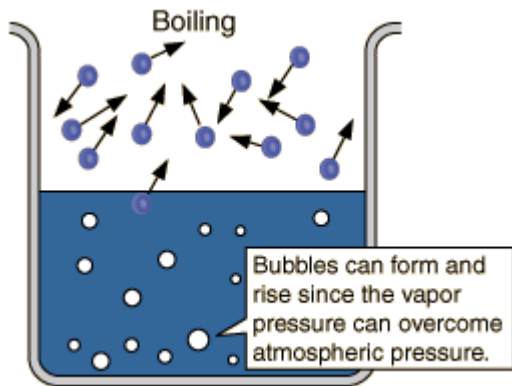
Evaporation: on the surface of condensed phase

Vapour pressure: equilibrium

$$N_{\text{escaping}} = N_{\text{returning}}$$

Boiling temperature (T_b)

The temperature at which the vapour pressure of a liquid is equal to the external pressure, vaporization can occur throughout the bulk (volume) of the liquid.

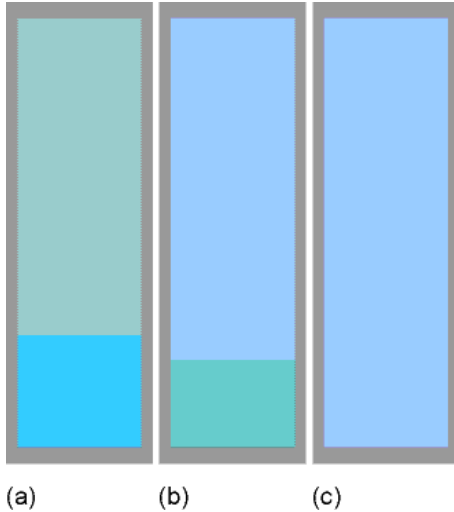


$p = 1 \text{ atm}$

The normal boiling point



3. Critical points



In a closed vessel

- (a) In equilibrium (vapour pressure).
- (b) With heating, the density of the vapour phase increases and that of the liquid decreases slightly.
- (c) There comes a stage, at which the two densities are equal and the l-g interface disappears.

Critical temperature (T_c)

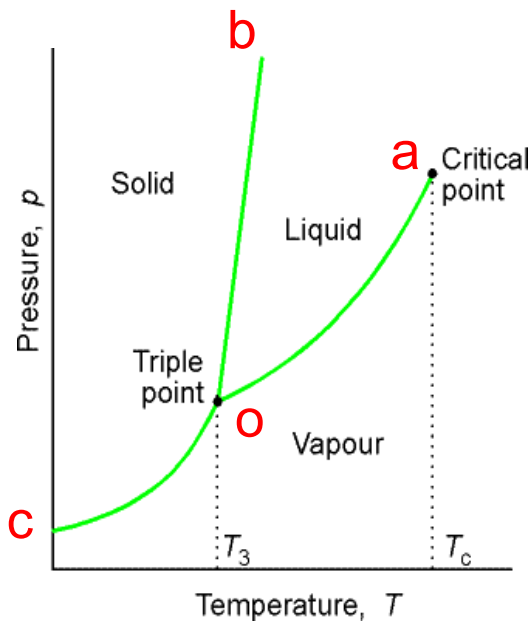
The temperature at which the surface disappears.

Critical pressure (p_c)

The vapour pressure at T_c

Supercritical fluid

The single uniform phase at T_c, p_c .





4. Melting points and triple points

Melting/freezing temperature (T_f)

Under a specified pressure, the temperature at which the liquid and solid phases of a substance coexist in equilibrium.

$p = 1 \text{ atm}$

The normal freezing point

Triple points

the **triple point** of a substance is the temperature and pressure at which the three phases (g,l,s) of that substance coexist in equilibrium, the point at which the three phase boundaries meet.

T_3 of a pure substance is a characteristic, unchangeable.



4. 3 Examples of typical phase diagrams

1. Water

✓ Phase transitions

✓ Four points

Normal $T_b = 373.15 \text{ K}$ ($p = 1 \text{ atm}$)

Normal $T_f = 273.15 \text{ K}$ ($p = 1 \text{ atm}$)

$T_c = 674.30 \text{ K}$ ($p = 218.3 \text{ atm}$)

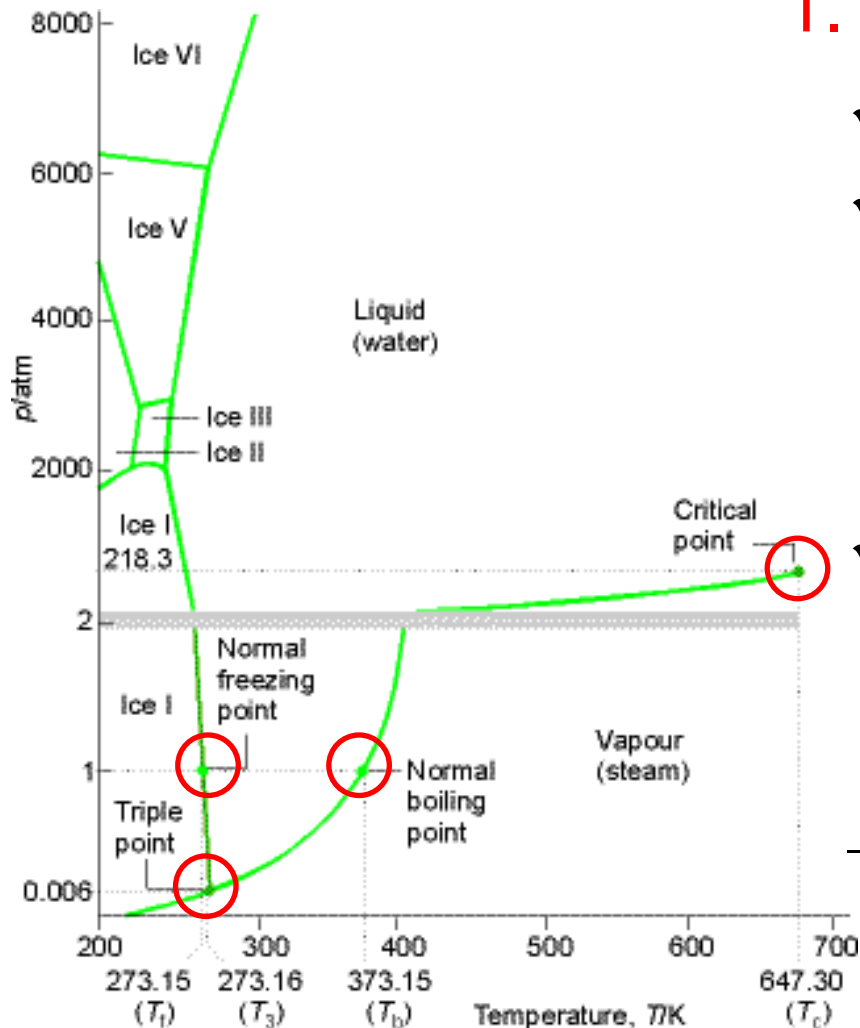
$T_3 = 273.16 \text{ K}$ ($p = 0.006 \text{ atm}$)

✓ Information

$p < 2 \text{ k atm}$

$p \uparrow$ $T_f \downarrow$ liquid phase favorable

For water, liquid is denser than the solid.





2. Carbon dioxide

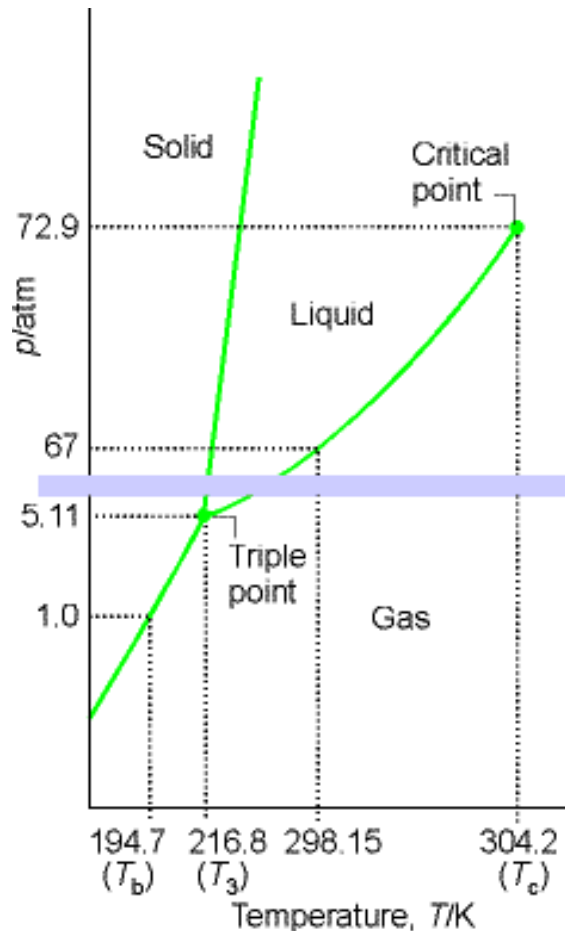
Information

The melting temperature of CO_2 (s) rises as the pressure is increased.

$$\rho_s > \rho_{l,g}$$

T_3 lies above 1 atm, the l phase cannot exist at normal atmospheric pressures whatever T

At 1 atm, the solid (dry ice) sublimates directly.





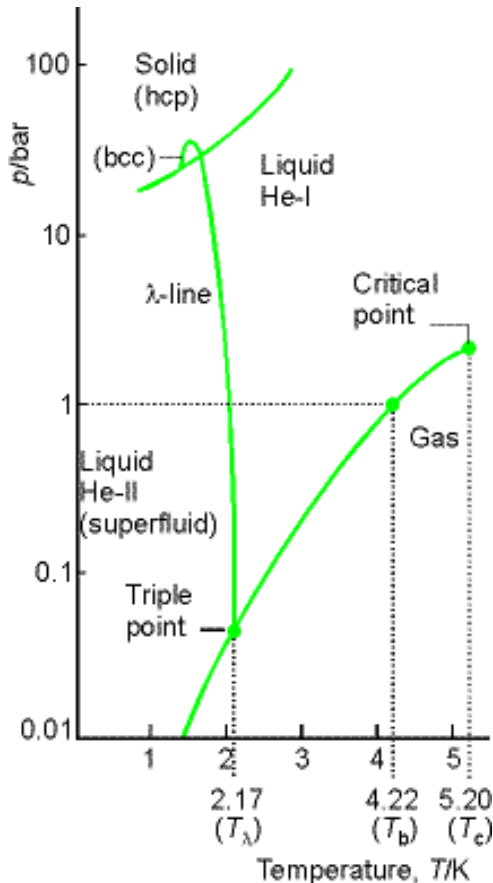
3. Helium (^4He)

Information

- Helium behaves unusually at low T
- The s-g phases are never in equilibrium at whatever T.
- Solid phase only at high pressure over 20 bar.
- The pure helium has a l-l phase transition at its λ -line.

He-I phase: normal liquid;

He-II phase: superfluid (without viscosity)





4.4 The dependence of stability on the conditions

1. Phase stability $\sim T$

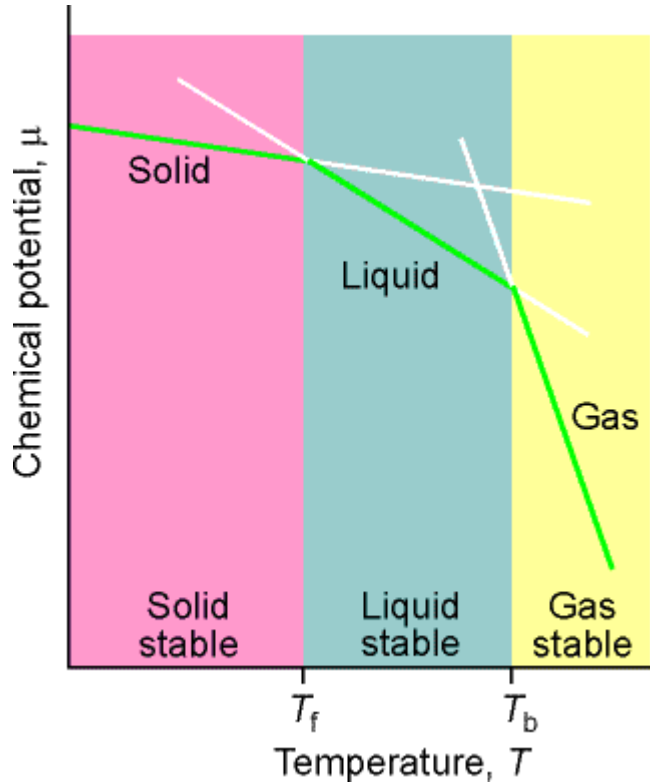
Phase stability criterion: μ chemical potential

Phase transition from high potential to low potential

Phase equilibrium when $\mu^\alpha = \mu^\beta$

Pure substance, constant p

$$\left(\frac{\partial \mu}{\partial T}\right)_p = -S_m \quad S_m > 0$$



$$\left(\frac{\partial \mu}{\partial T}\right)_p = -S_m$$

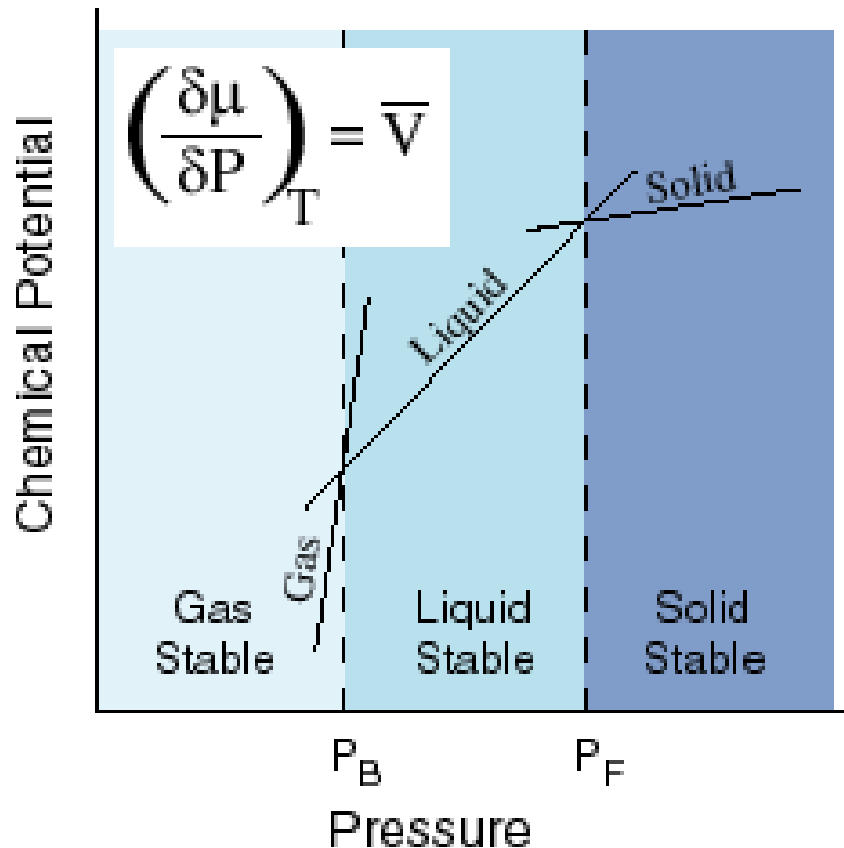
- In a give phase, $T \uparrow \mu \downarrow$
- $S_m(g) > S_m(l) > S_m(s)$
Same ΔT ,
 $\Delta \mu(g) > \Delta \mu(l) > \Delta \mu(s)$
- Constant p , $T \uparrow$
Phase transfer direction
Solid $\rightarrow$ liquid $\rightarrow$ gas



2. Phase stability ~ p

Pure substance, constant T

$$\left(\frac{\partial \mu}{\partial p}\right)_T = V_m \quad V_m > 0$$



- In a give phase, $p \uparrow \mu \uparrow$
- $V_m(g) > V_m(l) > V_m(s)$
Same Δp ,
 $\Delta \mu(g) > \Delta \mu(l) > \Delta \mu(s)$
- Constant T, $p \downarrow$
Phase transfer direction
Solid $\rightarrow$ liquid $\rightarrow$ gas



Example

Calculate the effect on the chemical potentials of ice and water of increasing the pressure from 1.00 bar to 2.00 bar at 0°C. The density of ice is 0.917 gcm⁻³ and that of liquid water is 0.999 gcm⁻³ under these conditions.

$$T = 273 \text{ K}$$

$$\Delta p = 2 \text{ bar} - 1 \text{ bar} = 1 \text{ bar}$$

$$\rho (\text{s}) = 0.917 \text{ gcm}^{-3} ; \rho (\text{l}) = 0.999 \text{ gcm}^{-3}$$

$$\left(\frac{\partial \mu}{\partial p} \right)_T = V_m$$



We know that the change in chemical potential of an incompressible substance when the pressure is change by Δp is:

$$\Delta\mu = V_m\Delta p$$

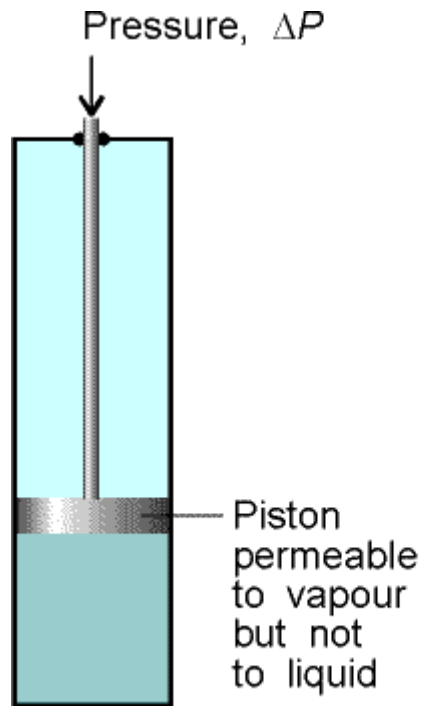
$$V_m = M/\rho$$

$$M(\text{H}_2\text{O}) = 18 \text{ g mol}^{-1}$$

$$\begin{aligned}\Delta\mu(\text{ice}) &= \frac{M\Delta p}{\rho} = \frac{(0.018 \text{ kg mol}^{-1}) \times (1 \times 10^5 \text{ Pa})}{917 \text{ kg m}^{-3}} \\ &= 1.97 \text{ J mol}^{-1}\end{aligned}$$

$$\begin{aligned}\Delta\mu(\text{water}) &= \frac{M\Delta p}{\rho} = \frac{(0.018 \text{ kg mol}^{-1}) \times (1 \times 10^5 \text{ Pa})}{999 \text{ kg m}^{-3}} \\ &= 1.80 \text{ J mol}^{-1}\end{aligned}$$

3. The effect of applied pressure on vapour pressure



At constant T , an additional pressure ΔP was applied to liquid surface. If the vapour pressure of the liquid in the absence of an additional pressure is p^* , calculate the new vapour pressure p in equilibrium?

$$\mu(l) = \mu(g)$$

$$d\mu(l) = d\mu(g)$$

$$d\mu(l) = V_m(l)dP$$

$$d\mu(g) = V_m(g)dp = \frac{RT}{p}dp$$

$$\frac{RT}{p}dp = V_m(l)dP$$

Gas phase

initial p^* ; final p

Liquid phase

initial p^* ; final $p+\Delta P$



$$\frac{RT}{p} dp = V_m(l) dP$$

Assume that $V_m(l)$ of the liquid is the same throughout the small range of pressures involved:

$$p \approx p^*$$

$$RT \int_{p^*}^p \frac{dp}{p} = \int_{p^*}^{p+\Delta P} V_m(l) dP$$

$$RT \ln \frac{p}{p^*} = V_m(l) \Delta P$$

$$p = p^* e^{V_m \Delta P / RT}$$

$$e^x = 1 + x + \frac{1}{2}x^2 + \dots$$

$$\text{If } x \ll 1$$

$$e^x = 1 + x$$

if

$$\frac{V_m \Delta P}{RT} \ll 1$$

$$p \approx p^* \left(1 + \frac{V_m \Delta P}{RT} \right)$$

$$\frac{p - p^*}{p^*} \approx \frac{V_m \Delta P}{RT}$$

外压变化和蒸汽压的关系



Calculate the fractional increase of the vapour pressure of water for an increase in pressure of 10 bar at 25 °C.

$$\Delta P = 1 \times 10^6 \text{ Pa}$$

$$T = 298 \text{ K, for water, } M = 0.18 \text{ kg/mol; } \rho = 999 \text{ kg m}^{-3}$$

$$\frac{V_m \Delta P}{RT} = \frac{M \Delta P}{\rho RT}$$

$$p = p^* e^{V_m \Delta P / RT}$$

$$\frac{p - p^*}{p^*} \approx \frac{V_m \Delta P}{RT}$$

$$= \frac{(0.018 \text{ kg mol}^{-1}) \times (1 \times 10^6 \text{ Pa})}{(999 \text{ kg m}^{-3}) \times (8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times 298 \text{ K}}$$

$$= 7.3 \times 10^{-3}$$



Calculate the fractional increase of the vapour pressure of water for an increase in pressure of 10 bar at 25 °C.

$$\therefore \frac{V_m \Delta P}{RT} = 7.3 \times 10^{-3} \ll 1$$

$\therefore$

$$\frac{p - p^*}{p^*} \approx \frac{V_m \Delta P}{RT}$$

$$\frac{p - p^*}{p^*} = 7.3 \times 10^{-3}$$

An increase of 0.73 per cent.

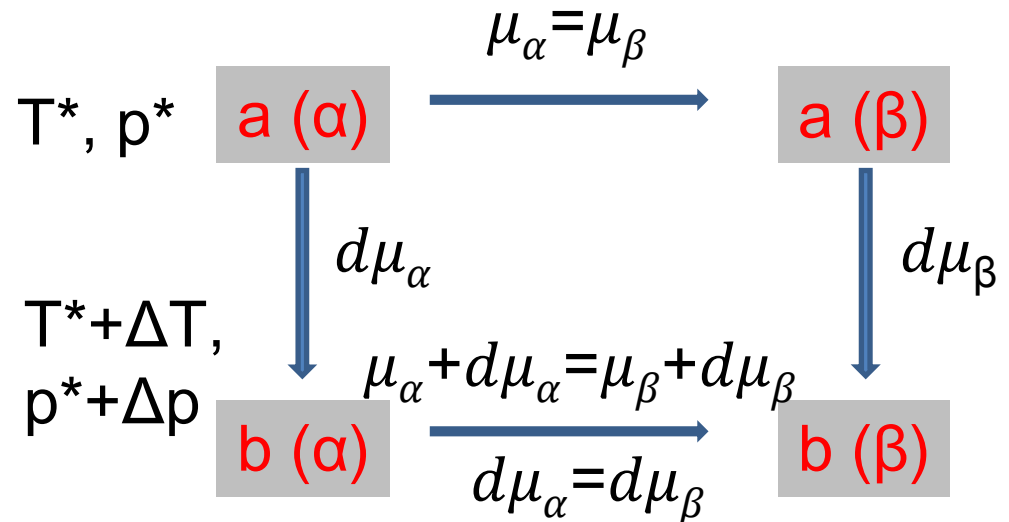
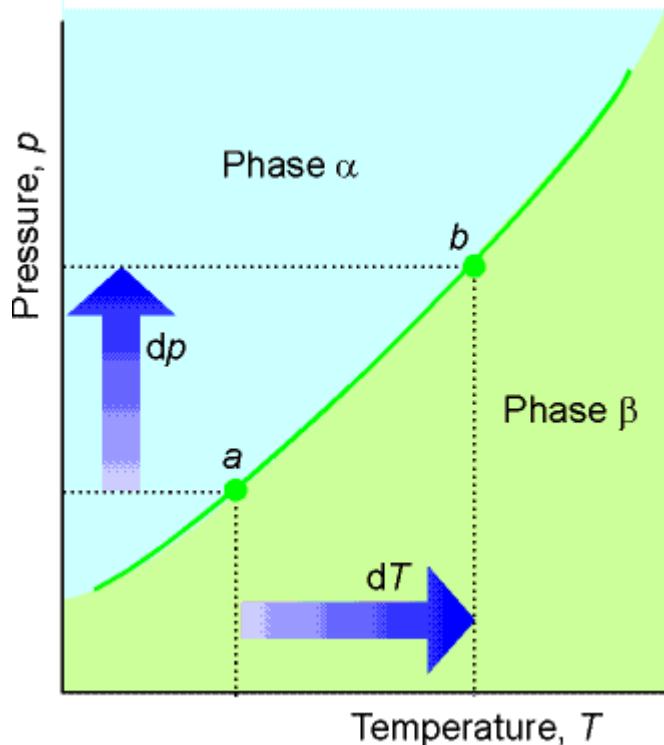


4.5 The location of phase boundaries

Phases are in equilibrium

$$\mu_{\alpha}(T, p) = \mu_{\beta}(T, p)$$

1. The slopes of the phase boundaries



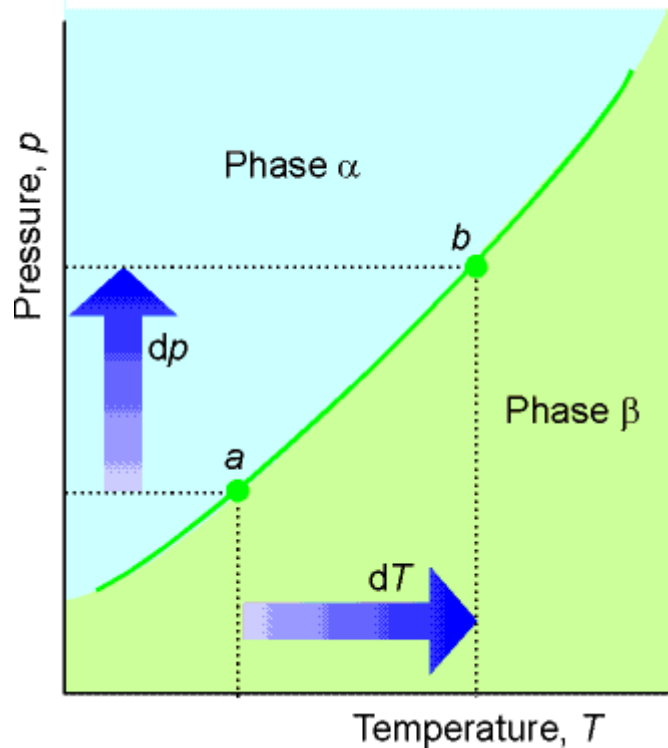


$$d\mu = dG_m = -S_m dT + V_m dp$$

$$\begin{aligned} -S_{\alpha,m} dT + V_{\alpha,m} dp \\ = -S_{\beta,m} dT + V_{\beta,m} dp \end{aligned}$$

$$(V_{\beta,m} - V_{\alpha,m}) dp = (S_{\beta,m} - S_{\alpha,m}) dT$$

$$\begin{aligned} \frac{dp}{dT} &= \frac{S_{\beta,m} - S_{\alpha,m}}{V_{\beta,m} - V_{\alpha,m}} \\ &= \frac{\Delta_{trs} S}{\Delta_{trs} V} = \frac{\Delta_{\beta \rightarrow \alpha} S}{\Delta_{\beta \rightarrow \alpha} V} \end{aligned}$$





Clapeyron equation

$$\frac{dp}{dT} = \frac{\Delta_{trs}S}{\Delta_{trs}V}$$

- ① Phase equilibrium of any pure substance
- ② thermodynamic data → phase diagrams

Phase equilibrium

$$\Delta_{trs}S = \frac{\Delta_{trs}H}{T}$$

At a give pressure

T is the phase transition temperature

$$\frac{dp}{dT} = \frac{\Delta_{trs}S}{\Delta_{trs}V} = \frac{\Delta_{trs}H}{T\Delta_{trs}V}$$



2. The solid-liquid boundary

$$\frac{dp}{dT} = \frac{\Delta_{fus}H}{T\Delta_{fus}V}$$

Generally, $\Delta_{fus}H > 0$ (melting is endothermic)

$\Delta_{fus}V > 0$ (s-l thermal expansion is small)

Conclusion: The slope of the solid-liquid boundary is steep and usually positive.



3. The liquid-vapour boundary

$$\frac{dp}{dT} = \frac{\Delta_{vap}H}{T\Delta_{vap}V}$$

Generally, $\Delta_{vap}H > 0$ (evaporation is endothermic)
 $\Delta_{vap}V > 0$ (l-g thermal expansion is big)

Conclusion: The slope of the liquid-vapour boundary is positive but less steep than that for the solid-liquid boundary.



4. The solid-vapour boundary

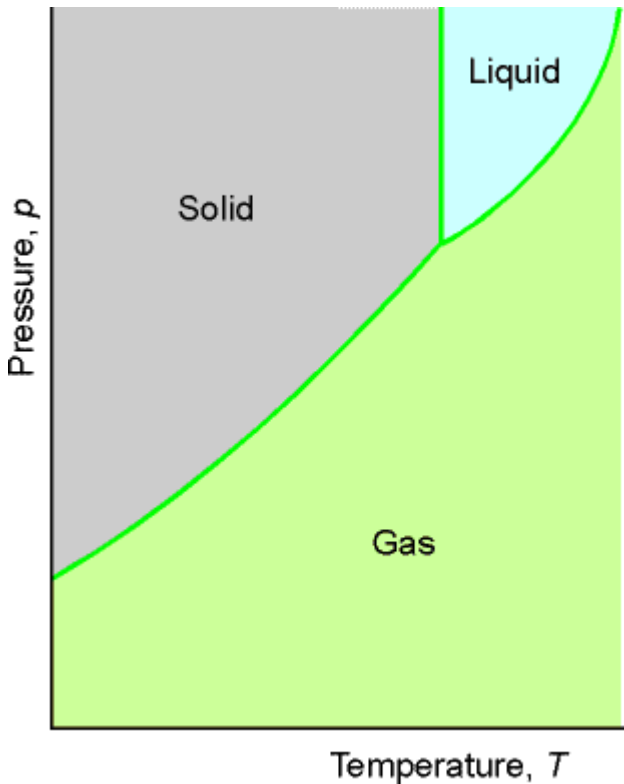
$$\frac{dp}{dT} = \frac{\Delta_{sub}H}{T\Delta_{sub}V}$$

Generally,

$\Delta_{sub}H > 0$ (sublimation is endothermic)

$\Delta_{sub}H > \Delta_{vap}H$

$\Delta_{sub}V > 0$ (s-g thermal expansion is big)



$$\left(\frac{dp}{dT}\right)_{s-l}$$

steepest

$$\left(\frac{dp}{dT}\right)_{s-g} > \left(\frac{dp}{dT}\right)_{l-g}$$



Example

Estimate the typical size of the effect of increasing pressure on the boiling point of a liquid.

$$\frac{dp}{dT} = \frac{\Delta_{vap}H}{T\Delta_{vap}V}$$

At the boiling point, the term $\frac{\Delta_{vap}H}{T}$ is Trouton's constant (Section 4.3a).

$$\Delta_{vap}V = V_m(g) - V_m(l) \approx V_m(g)$$



Example

Estimate the typical size of the effect of increasing pressure on the boiling point of a liquid.

Trouton's constant $\frac{\Delta_{vap}H}{T} = 85 \text{ JK}^{-1}\text{mol}^{-1}$.

$V_m(g) = 25 \text{ L mol}^{-1}$ at 1 atm and near but above T_{room} .

$$\frac{dp}{dT} \approx \frac{\Delta_{vap}H}{T\Delta_{vap}V} = \frac{85 \text{ JK}^{-1}\text{mol}^{-1}}{25 \times 10^{-3} \text{ m}^3\text{mol}^{-1}} = 3.4 \times 10^3 \text{ PaK}^{-1}$$

This value corresponds to 0.034 atm K^{-1} , and hence to $\frac{dT}{dp} = 30 \text{ K atm}^{-1}$. Therefore, a change of $+0.1 \text{ atm}$ can be expected to change a T_b by about $+3 \text{ K}$.



$$\because V_m(g) \gg V_m(l)$$

$$\because \Delta_{vap}V = V_m(g) - V_m(l) \approx V_m(g)$$

$$V_m(g) = RT/p$$

$$\frac{dp}{dT} = \frac{\Delta_{vap}H}{T\Delta_{vap}V}$$

$$\frac{dp}{pdT} = \frac{\Delta_{vap}H}{RT^2}$$

$$dx/x = d \ln x$$

$$\frac{d \ln p}{dT} = \frac{\Delta_{vap}H}{RT^2}$$



$$d \ln p = \frac{\Delta_{vap}H}{RT^2} dT$$

$$\int_{p^*}^p d \ln p = \int_{T^*}^T \frac{\Delta_{vap}H}{RT^2} dT$$

$$\int \frac{1}{x^2} dx = -\frac{1}{x}$$

$$\ln p - \ln p^* = \ln \frac{p}{p^*} = -\frac{\Delta_{vap}H}{R} \left(\frac{1}{T} - \frac{1}{T^*} \right)$$

$$p = p^* e^{-\chi}$$

$$\chi = \frac{\Delta_{vap}H}{R} \left(\frac{1}{T} - \frac{1}{T^*} \right)$$

已知一个温度下的蒸汽压求另外一个温度下的蒸汽压（温度和蒸汽压的关系）



$$p = p^* e^{-\chi}$$

$$\chi = \frac{\Delta_{vap}H}{R} \left(\frac{1}{T} - \frac{1}{T^*} \right)$$

Estimate the vapour pressure of liquid at any temperature from its normal boiling point at 1 atm.

T_b of benzene at 1 atm is 80 °C (353 K),

$$\Delta_{vap}H^\theta = 30.8 \text{ kJ mol}^{-1}$$

Calculate the vapour pressure at 20 °C (293 K)

$$\chi = \frac{3.08 \times 10^4 \text{ J mol}^{-1}}{8.3145 \text{ J K}^{-1} \text{ mol}^{-1}} \left(\frac{1}{293 \text{ K}} - \frac{1}{353 \text{ K}} \right)$$

$$p^* = 1 \text{ atm}$$

$$p = p^* e^{-\chi} = 0.117 \text{ atm}$$



4.6 The classification of phase transition

Ehrenfest classification

There are many different types of phase transition, and **thermodynamic properties of substances** may be used to classify phase transitions into different types.

The first/second derivative of the chemical potential with respect to temperature.

$$\left(\frac{\partial \mu}{\partial T}\right)_p = -S_m$$

$$\left(\frac{\partial \left(\frac{\partial \mu}{\partial T}\right)_p}{\partial T}\right)_p = \left(\frac{\partial (-S_m)}{\partial T}\right)_p = -\frac{C_{p,m}}{T}$$



Ehrenfest's Classification

- ✓ First order phase transition: Discontinuity in the first derivative of Gibb's Free Energy, G .
- ✓ Second order phase transition: Continuous first derivative but discontinuity in the second derivative of G .

Higher-order phase transitions

	Differential of G		Discontinuity appears in: Corresponding experim. quantities		
	S	V	S	V	
First					
Second	$\left(\frac{\partial S}{\partial T}\right)_P$	$\left(\frac{\partial V}{\partial T}\right)_P$	C_P	β	κ
	$\left(\frac{\partial S}{\partial p}\right)_T$	$\left(\frac{\partial V}{\partial p}\right)_T$			



1. first-order phase transition

The first derivative of the chemical potential with respect to temperature and pressure is discontinuous.

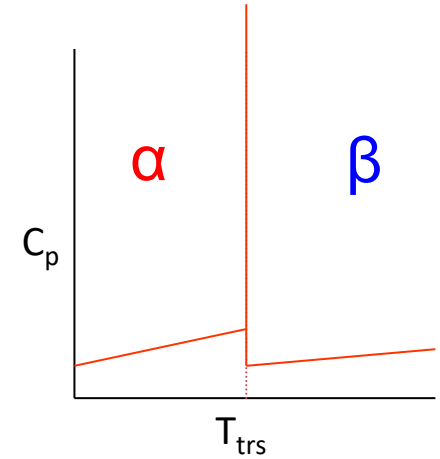
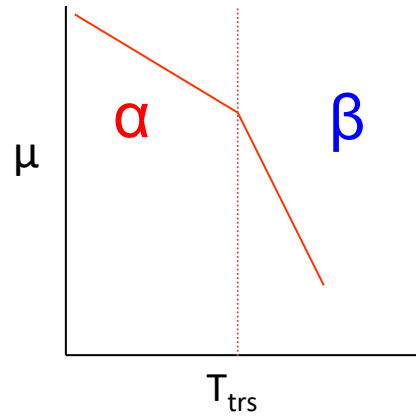
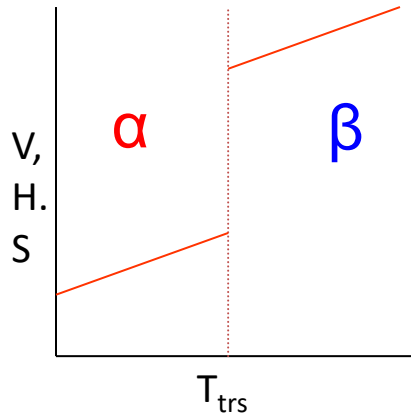
At transition from a phase, α , to another phase, β . Transitions are accompanied by changes in entropy and volume.

$$\begin{aligned}\left(\frac{\partial \mu_{\beta}}{\partial T}\right)_p - \left(\frac{\partial \mu_{\alpha}}{\partial T}\right)_p &= -S_{\beta,m} - (-S_{\alpha,m}) \\ &= -\Delta S_{trs} = -\frac{\Delta H_{trs}}{T}\end{aligned}$$

$$\left(\frac{\partial \mu_{\beta}}{\partial p}\right)_T - \left(\frac{\partial \mu_{\alpha}}{\partial p}\right)_T = V_{\beta,m} - V_{\alpha,m} = \Delta V_{trs}$$



1st phase transition



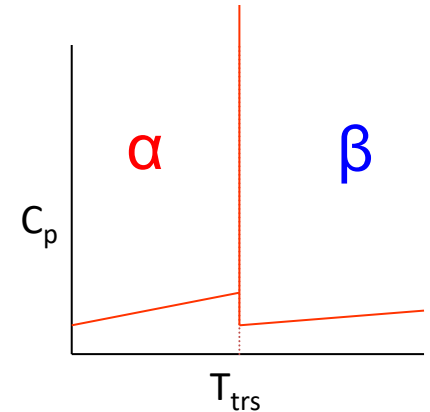
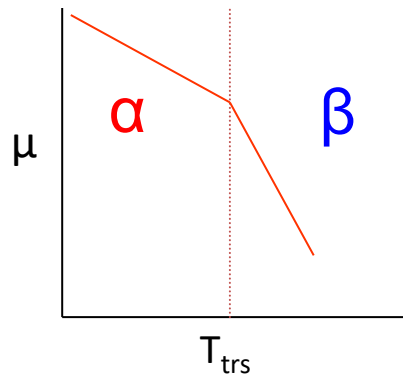
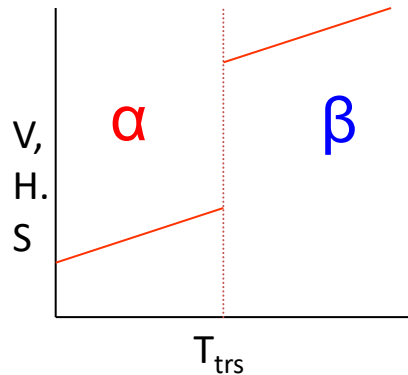
At equilibrium,

$$\mu(\alpha) = \mu(\beta), \Delta G = 0,$$

$$\text{but, } \Delta H \neq 0, \Delta S \neq 0, \Delta V \neq 0.$$



1st phase transition



$$C_p = \left(\frac{\partial H}{\partial T} \right)_p$$

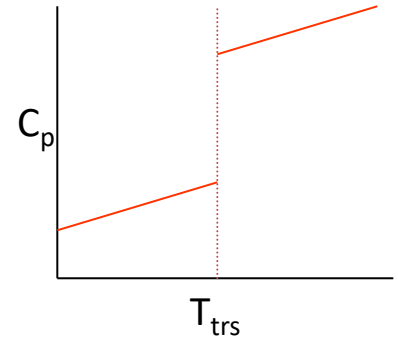
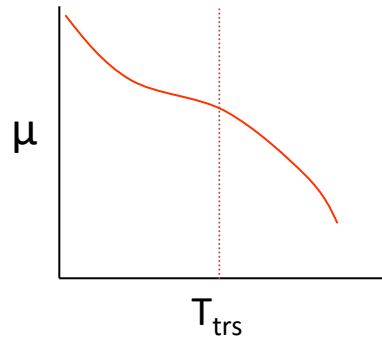
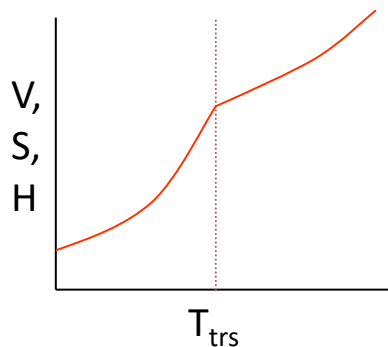
At a 1st phase transition, ΔH is a finite amount for an infinitesimal change of T . Therefore, C_p is infinite at the transition.

The physical reason is that heating drives the transition rather than raising the temperature.



2. Second-order phase transition

The first derivative of the chemical potential with respect to temperature is continuous but its second derivative is discontinuous..



At equilibrium,

$$\mu(\alpha) = \mu(\beta), \Delta G = 0,$$

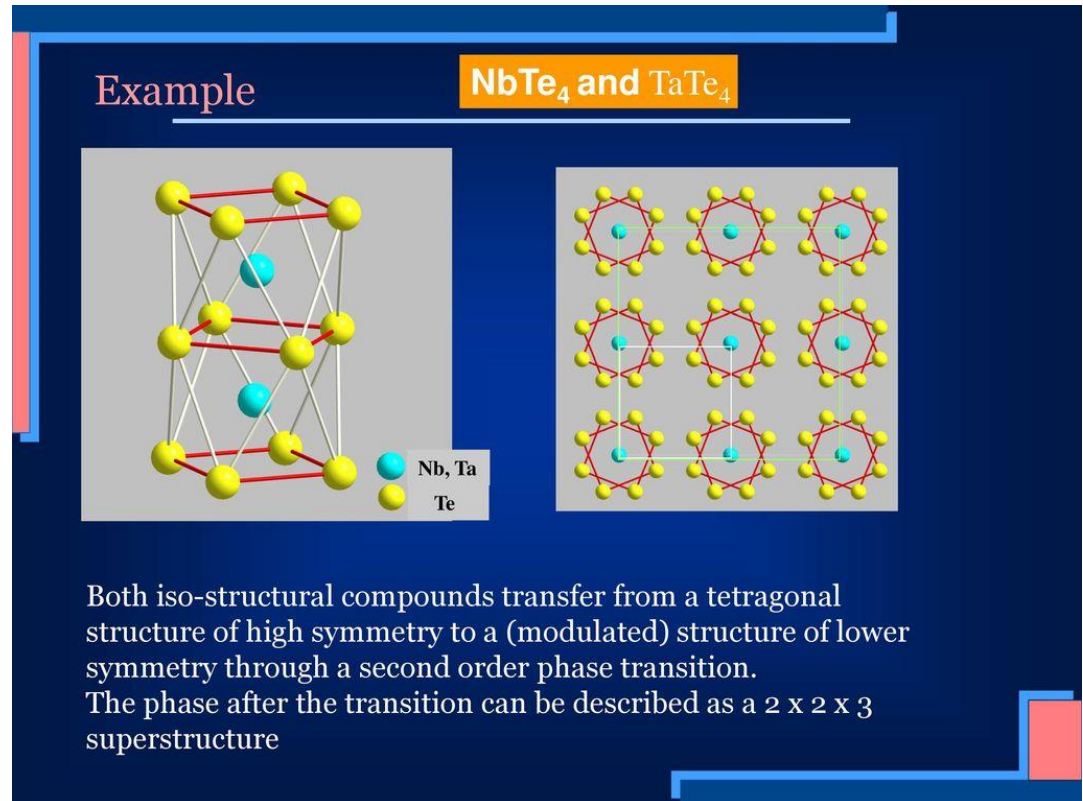
$$\text{but, } \Delta H = 0, \Delta S = 0, \Delta V = 0.$$

C_p is discontinuous but a finite value

2. Second-order phase transition

Example:

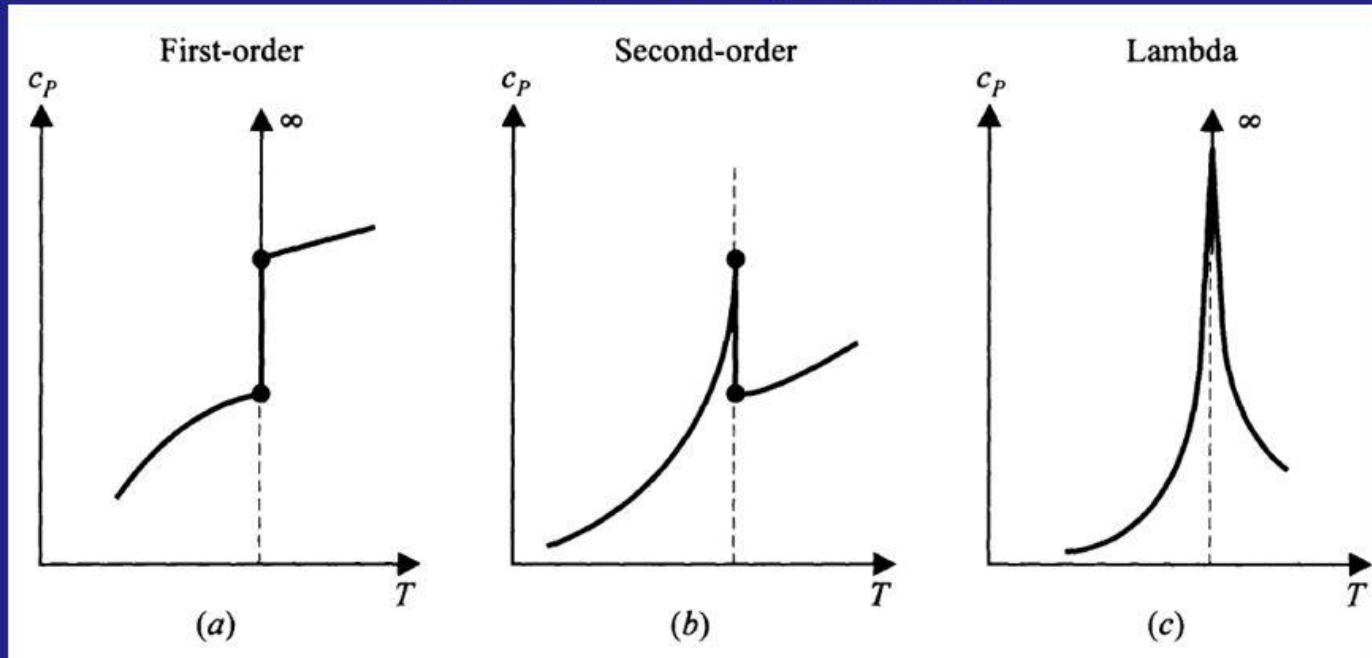
- Transition of a super conductor from super conducting to the normal state in zero magnetic field.
- Ferromagnetic to paramagnetic transition in a simple model.





1st-order, 2nd-order and Lambda-type transitions

$$[T_1 = T_2 ; P_1 = P_2 ; g_1 = g_2]$$



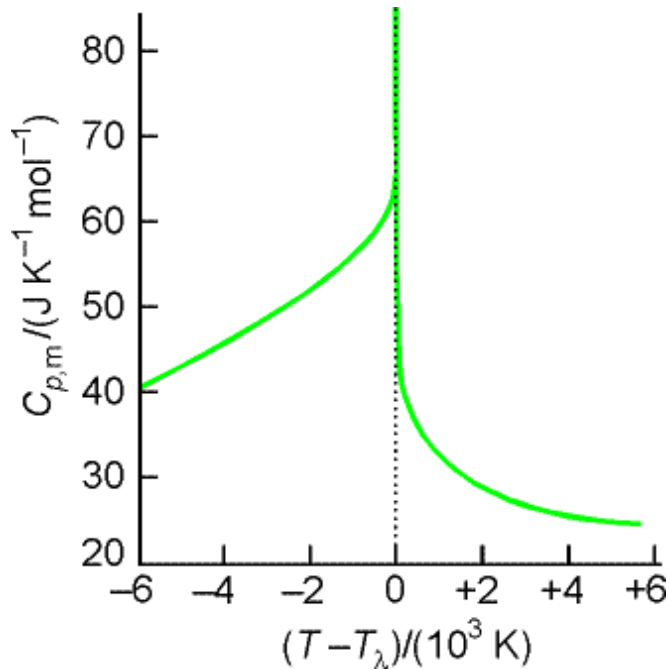
$$\begin{aligned} V_1 &\neq V_2 \\ S_1 &\neq S_2 \\ c_p, \alpha, \kappa_T &\rightarrow \infty \end{aligned}$$

$$\begin{aligned} V_1 &= V_2 \\ S_1 &= S_2 \\ c_{p1} &\neq c_{p2} \\ \alpha_1 &\neq \alpha_2 \\ \kappa_{T1} &\neq \kappa_{T2} \end{aligned}$$

$$\begin{aligned} V_1 &= V_2 \\ S_1 &= S_2 \\ c_p, \alpha, \kappa_T &\rightarrow \infty \end{aligned}$$



3. λ -transition



- ✓ Order-disorder transition in alloys
- ✓ The onset of ferromagnetism
- ✓ Fluid-superfluid transition of liquid He

The term λ -transition is applied to a phase transition that is not 1st order yet the C_p becomes infinite at the transition temperature.

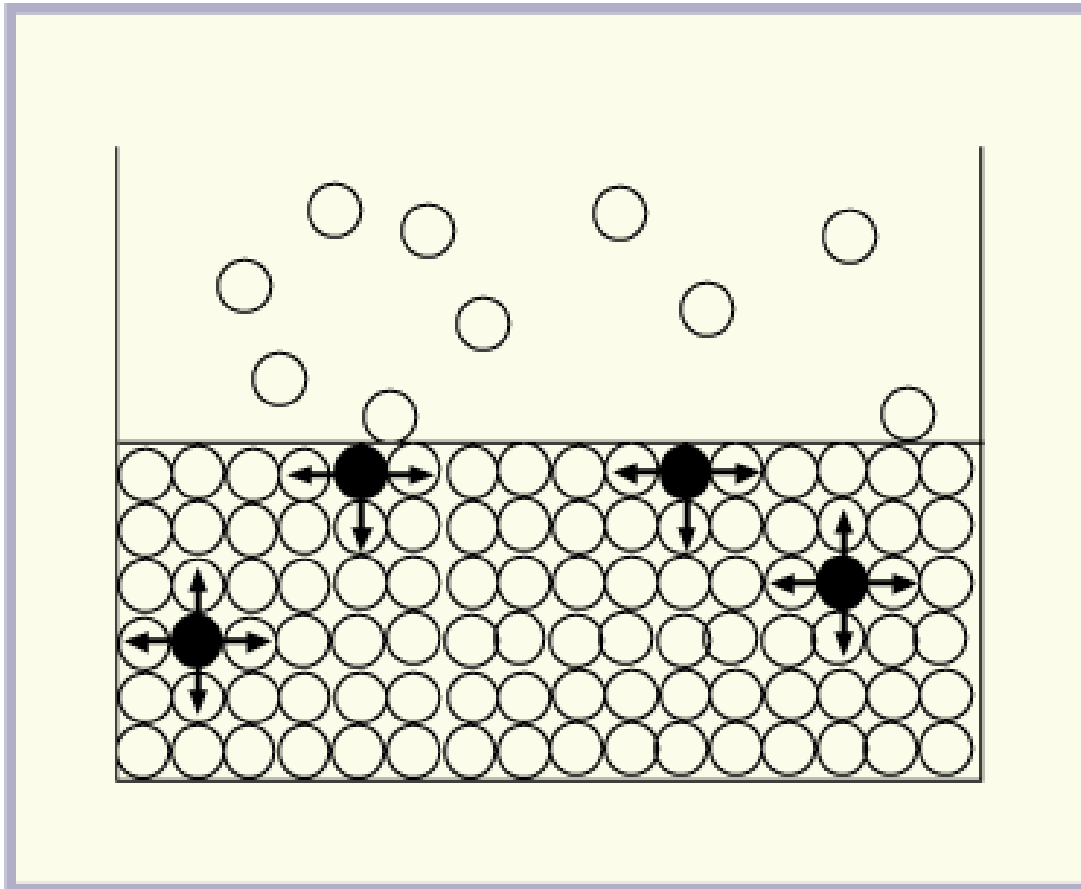
Typically, the C_p of a system that shows such a transition begins to increase well before the transition and the shape of the heat capacity curve resembles the Greek letter λ .





4.7 The physical liquid surface

Liquid-vapour interface





4.7 The physical liquid surface

Liquid-vapour interface

Liquids tend to adopt shapes that minimize their surface area, for then the maximum number of molecules are in the bulk and hence surrounded by and interacting with neighbors. Droplets of liquids therefore tend to be spherical, because a sphere is the shape with the smallest surface-to-volume ratio.



1. The surface tension

Surface effects may be expressed in terms of Helmholtz and Gibbs energies. and that dA and dG are equal to the work done in changing the energy of a system. The work needed to change the surface area, σ , of a sample by an infinitesimal amount $d\sigma$ is proportional to $d\sigma$:

$$dw = \gamma d\sigma$$

γ : surface tension

Unit: J m^{-2} or N m^{-1} ($1 \text{ J} = 1 \text{ N m}$)



The work of surface formation at **constant V and T** can be identified with the change in the Helmholtz energy (A):

$$dA = dw_{max} = dw' = \gamma d\sigma$$

$$\text{if } d\sigma < 0, dA < 0$$

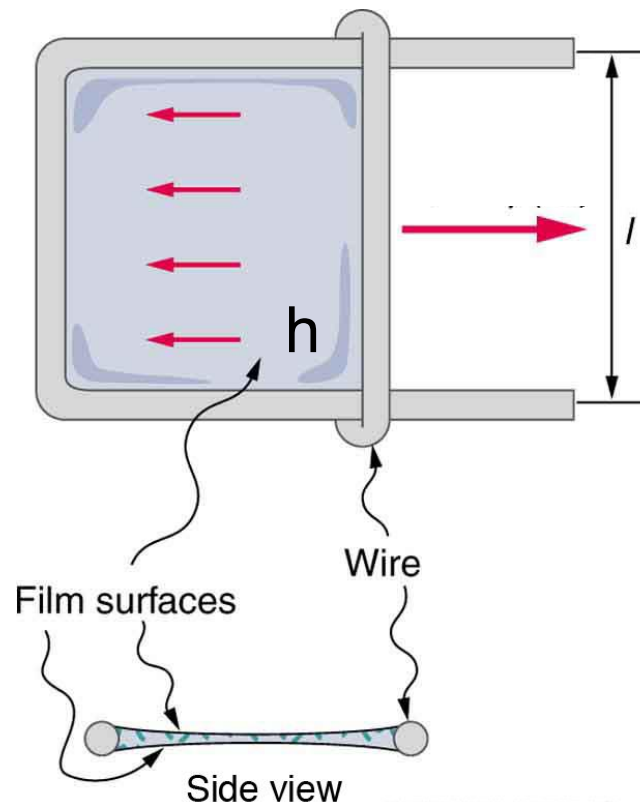
Spontaneous process

Surface have a natural tendency to contract.



Example

Calculate the work needed to raise a wire of length l and to stretch the surface of a liquid through a height h in the arrangement. Disregard gravitational potential energy.





The work required to create a surface area given that the surface tension does not vary as the surface is formed is $w = \gamma \sigma$. So, all we need do is to calculate the surface area of the two-sided rectangle formed as the frame is withdrawn from the liquid.

When the wire of length l is raised through a height h it increases the area of the liquid by twice the area of the rectangle.

The work done is:

$$w = \gamma \sigma = 2\gamma l h$$

$$w = \text{force} \times \text{distance} = 2\gamma l \times h$$



2. Curved surfaces

The minimization of the surface area of a liquid may result in the formation of a curved surface.



Bubble: a region in which vapour is trapped by a thin film.

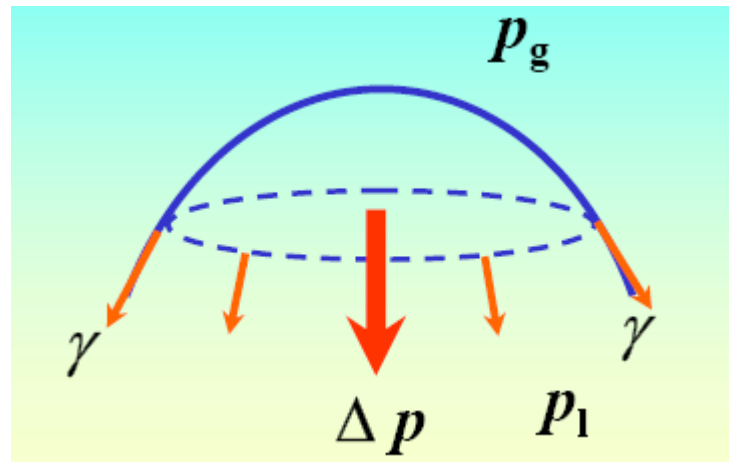
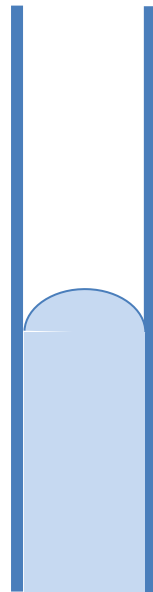
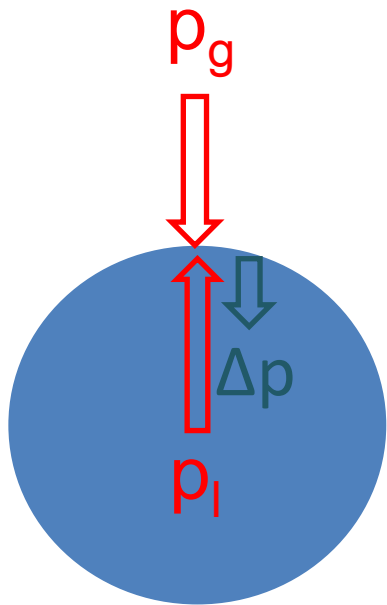
Cavity: a vapour-filled hole in a liquid.

Difference: a bubble has two liquid-vapour interfaces.
a cavity has one liquid-vapour interface.



Droplet: a small volume of liquid at equilibrium surrounded by its vapour.

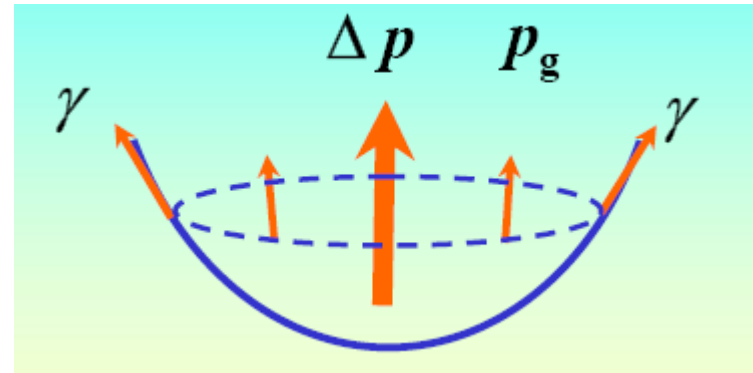
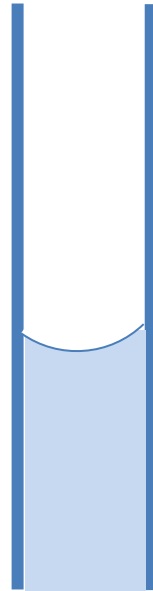
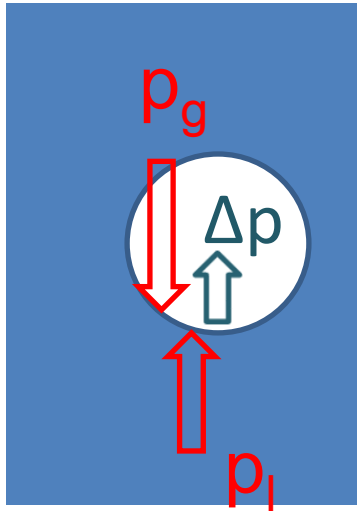
Laplace equation



$$P_l - P_g = \frac{2\gamma}{r} \quad P_{\text{in}} = P_{\text{out}} + \frac{2\gamma}{r}$$

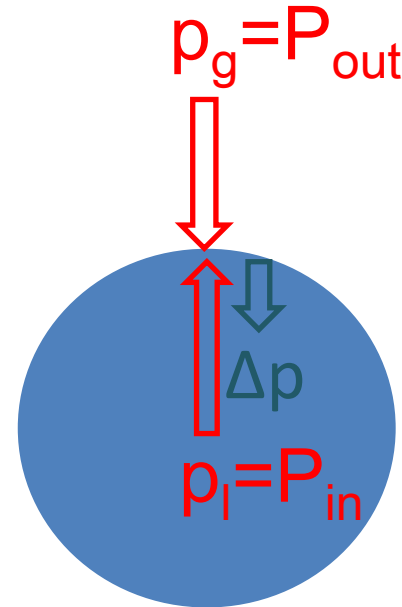
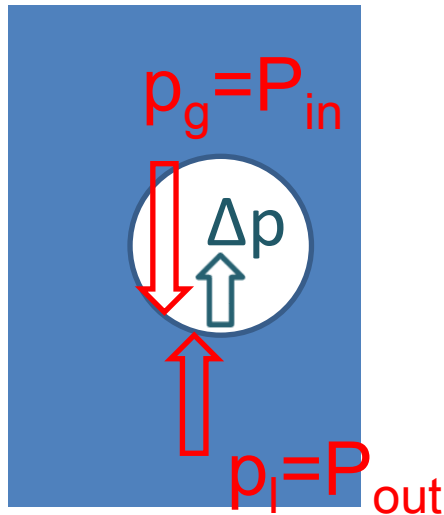


Laplace equation



$$P_g = P_l + \frac{2\gamma}{r} \quad P_{\text{in}} = P_{\text{out}} + \frac{2\gamma}{r}$$

$$P_l - P_g = -\frac{2\gamma}{r}$$



$$P_{in} = P_{out} + \left(\frac{2\gamma}{r} \right) \Rightarrow \text{Laplace pressure}$$



$$P_{in} = P_{out} + \Delta P$$

When the pressure inside a cavity is P_{in} and its radius is r , the outward force is

$$pressure \times area = 4\pi r^2 P_{in}$$

The force inwards F_2 arises from the external pressure $F_e = 4\pi r^2 P_{out}$ and the surface tension F_γ .



The radius of the cavity changes from r to $r+dr$

$$d\sigma = 4\pi(r + dr)^2 - 4\pi r^2 = 8\pi r dr$$

$$dw = 8\pi\gamma r dr = F_\gamma dr$$

$$F_\gamma = 8\pi\gamma r$$

$$\therefore 4\pi r^2 P_{in} = 4\pi r^2 P_{out} + 8\pi\gamma r$$

$$P_{in} = P_{out} + \frac{2\gamma}{r}$$

r smaller, ΔP more significant
e.g. if $r = 0.10$ mm, $\Delta P = 1.5$ kPa

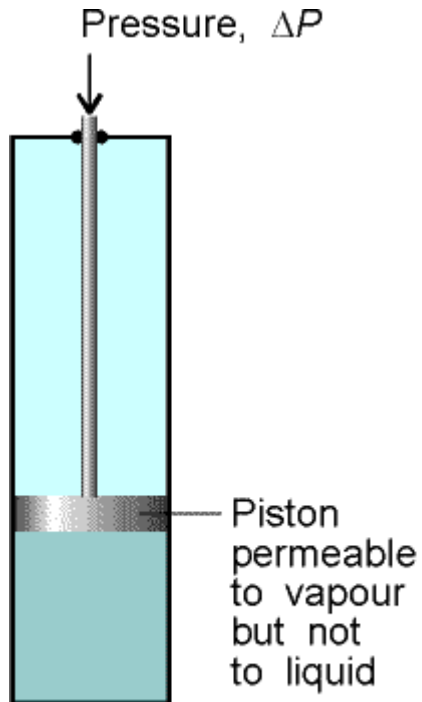
The effect of applied pressure on vapour pressure

$$p = p^* e^{V_m \Delta P / RT}$$

$$V_m = V_m(l) = \frac{M}{\rho}$$

$$p(l) - p(g) = \Delta P$$

ΔP Laplace pressure



Initial state is the flat liquid surface.
Final state is the curved liquid surface.
Calculate the vapour pressure for the curved liquid surface.



Kelvin equation

$$p = p^* e^{V_m \Delta P / RT}$$

For the vapour pressure of a liquid when it is dispersed as **droplets** of radius r .

Liquid in

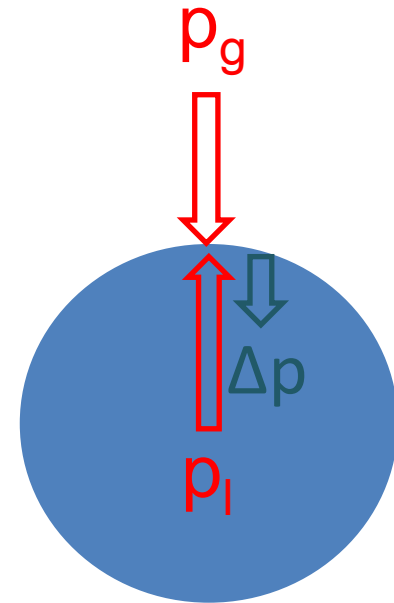
$$\Delta P = p(l) - p(g) = \frac{2\gamma}{r}$$

$$p = p^* e^{2\gamma V_m / rRT}$$

$$p > p^*$$

p^* : the vapour pressure of flat liquid

p : the vapour pressure of the curved liquid





Kelvin equation

The pressure of the liquid outside the cavity is less than the pressure inside, so for the vapour pressure inside a cavity:

Liquid out

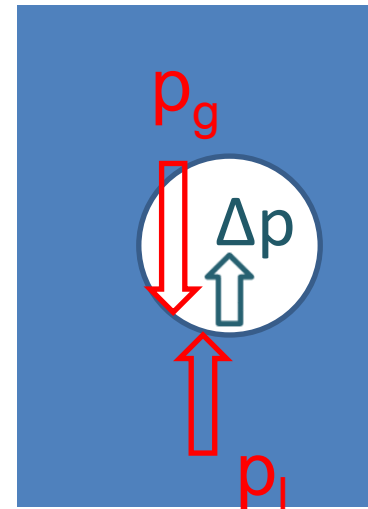
$$\Delta P = p(l) - p(g) = -\frac{2\gamma}{r}$$

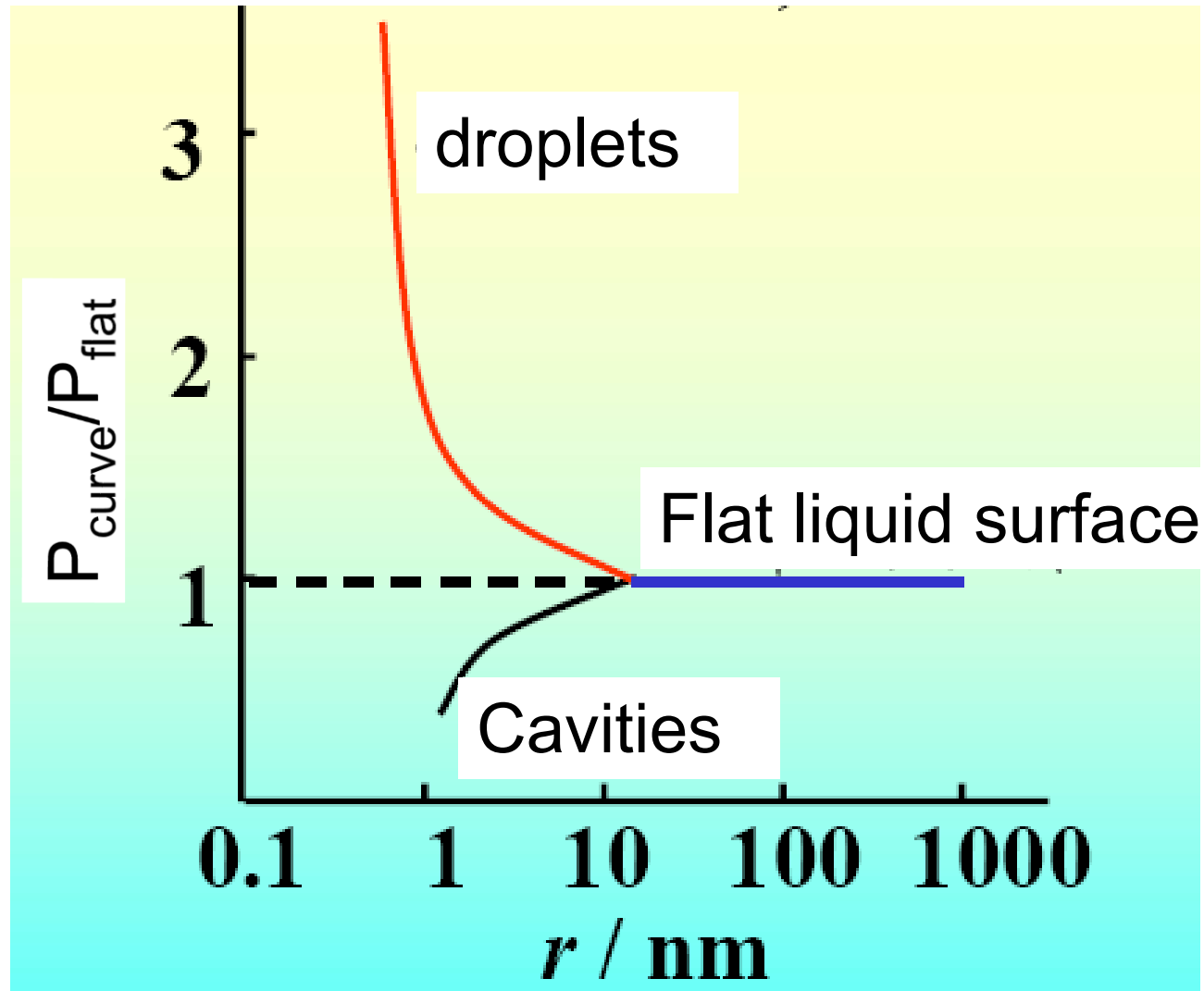
$$p = p^* e^{-2\gamma V_m / rRT}$$

$$p < p^*$$

p^* : the vapour pressure of flat liquid

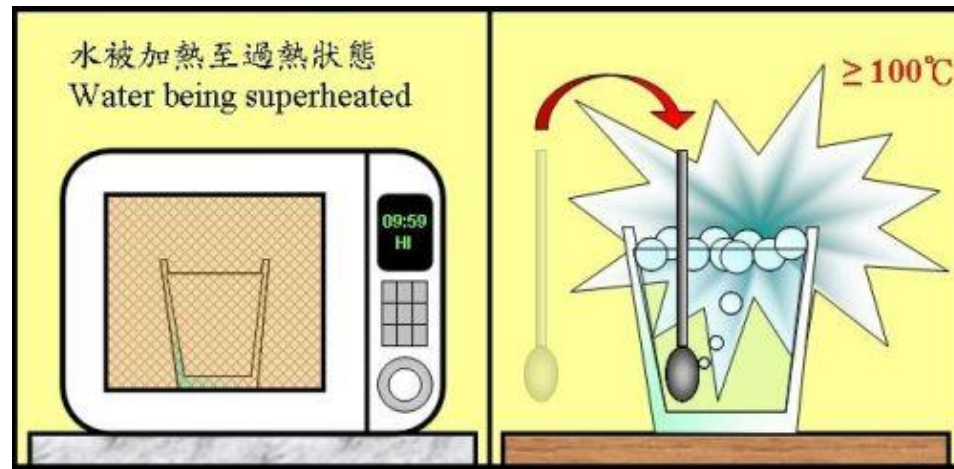
p : the vapour pressure of the curved liquid







Nucleation





$$\Delta P = \frac{2\gamma}{r}$$

$$p = p^* e^{2\gamma V_m / rRT}$$

Clouds forming: g to l phase transition

Initial droplets small so Kelvin equation tells us vapor pressure of droplet increases; Small droplets tend to evaporate; Air becomes **supersaturated** and thermodynamically unstable.

Supercooling-process of lowering the temperature of a liquid or a gas below its freezing point without it becoming a solid.

Unless large numbers of molecules congregate (spontaneous nucleation): Nucleation centers (dust particles).



$$\Delta P = \frac{2\gamma}{r}$$

$$p = p^* e^{-2\gamma V_m / rRT}$$

Superheating - liquids persist above boiling point

Vapor pressure inside a small cavity in liquid is low so cavities tend to collapse

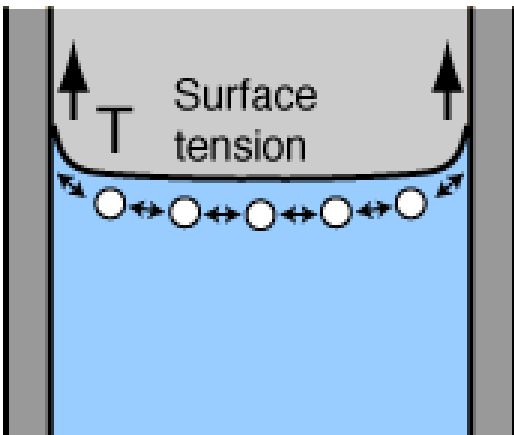
Spontaneous nucleation causes larger more stable cavities.

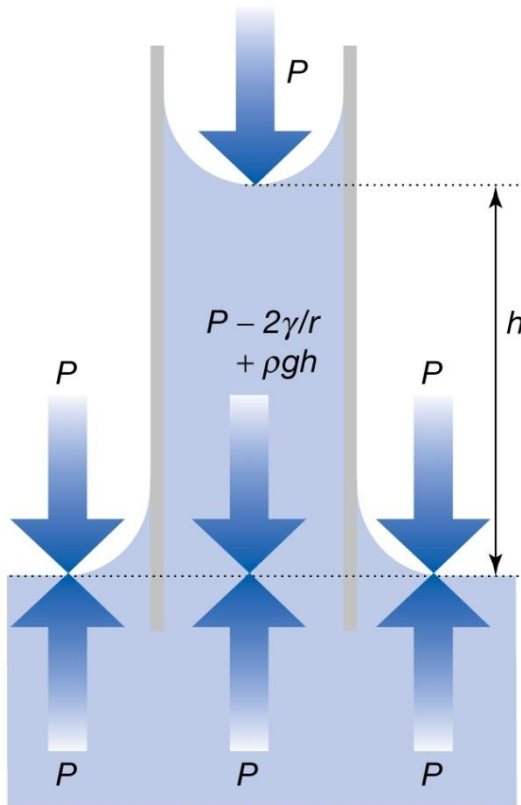
http://v.youku.com/v_show/id_XMTkyMzMxNjQ=.html?from=s1.8-1-1.2



3. Capillary action

Capillary action is the result of adhesion and surface tension. Adhesion of water to the walls of a vessel will cause an upward force on the liquid at the edges and result in a meniscus which turns upward. The surface tension acts to hold the surface intact, so instead of just the edges moving upward, the whole liquid surface is dragged upward.





Capillary action occurs when the adhesion to the walls is stronger than the cohesive forces between the liquid molecules.

The height **h** to which capillary action will lift water depends upon the weight of water which the surface tension will lift:

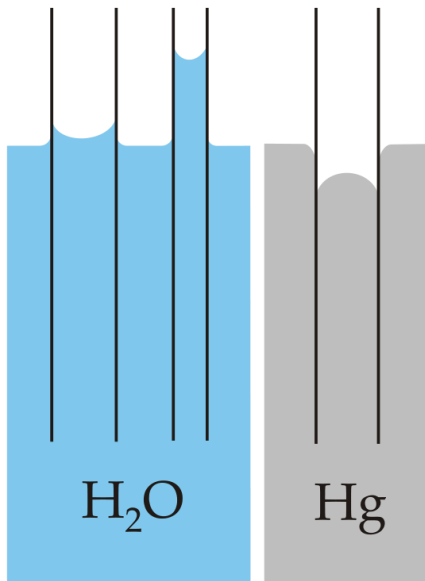
$$\rho gh = \frac{2\gamma}{r}$$

$$h = \frac{2\gamma}{\rho gr}$$



$$\gamma = \frac{hrg\rho^{(l)}}{2}$$

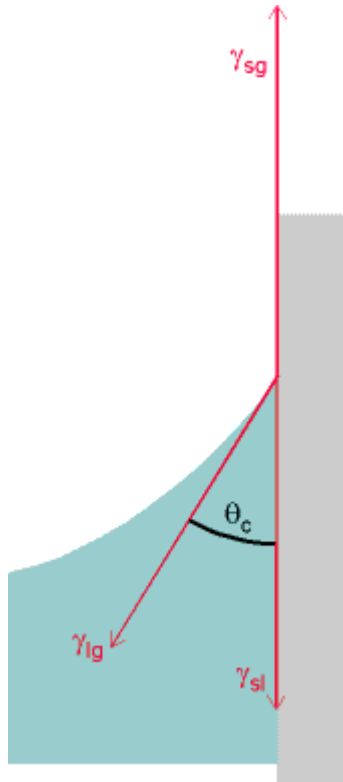
Application: measuring the surface tension of liquids.



When the adhesion to the walls is weaker than the cohesive forces between the liquid molecules (Hg), the liquid in the tube retracts from the walls.



4. Contact angle



The origin of the contact angle can be traced to the balance of forces at the line of contact between the liquid and the solid. If the solid-gas, solid-liquid, and liquid-gas surface tensions are denoted γ_{sg} , γ_{sl} and γ_{lg} , respectively, in balance:

$$\gamma_{sg} = \gamma_{sl} + \gamma_{lg} \cos \theta_c$$

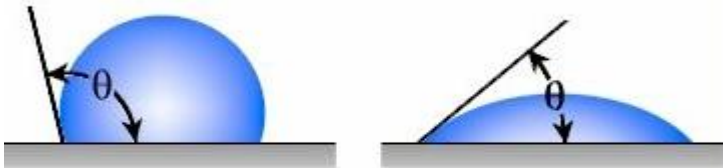
$$\cos \theta_c = \frac{\gamma_{sg} - \gamma_{sl}}{\gamma_{lg}}$$



Work of adhesion (w_{ad}) of liquid to solid is:

Hydrophobic Surface

Hydrophilic Surface



$$w_{ad} = \gamma_{sg} + \gamma_{lg} - \gamma_{sl}$$

$$\begin{aligned} \cos \theta_c &= (\gamma_{sg} - \gamma_{sl}) / \gamma_{lg} \\ &= (w_{ad} - \gamma_{lg}) / \gamma_{lg} \\ &= (w_{ad} / \gamma_{lg}) - 1 \end{aligned}$$

high
poor
poor
low

contact angle
adhesiveness
wettability
solid surface free
energy

low
good
good
high

If liquid wets surface

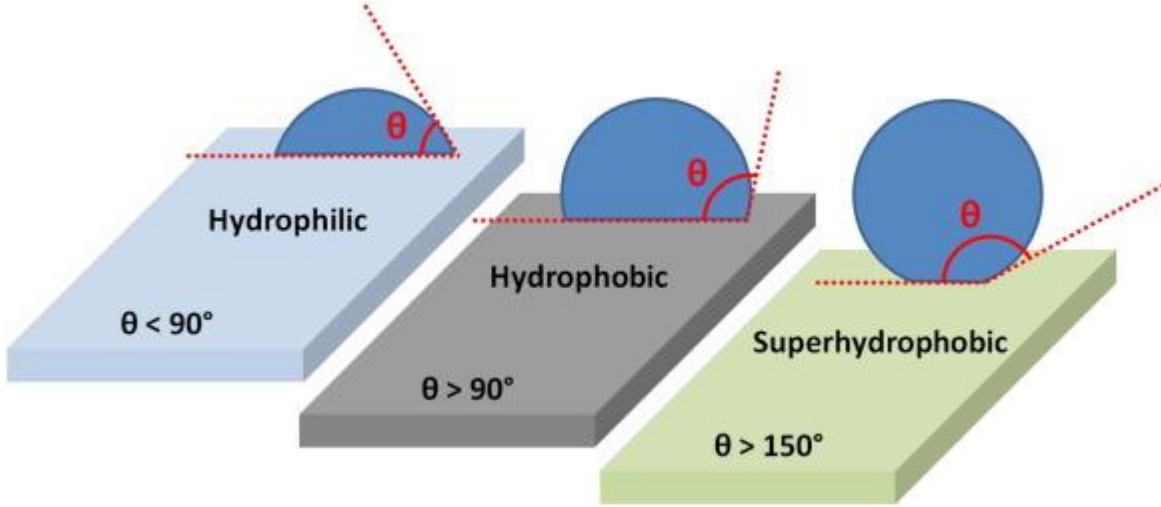
$$0^\circ < \theta_c < 90^\circ, 1 > \cos \theta_c > 0$$

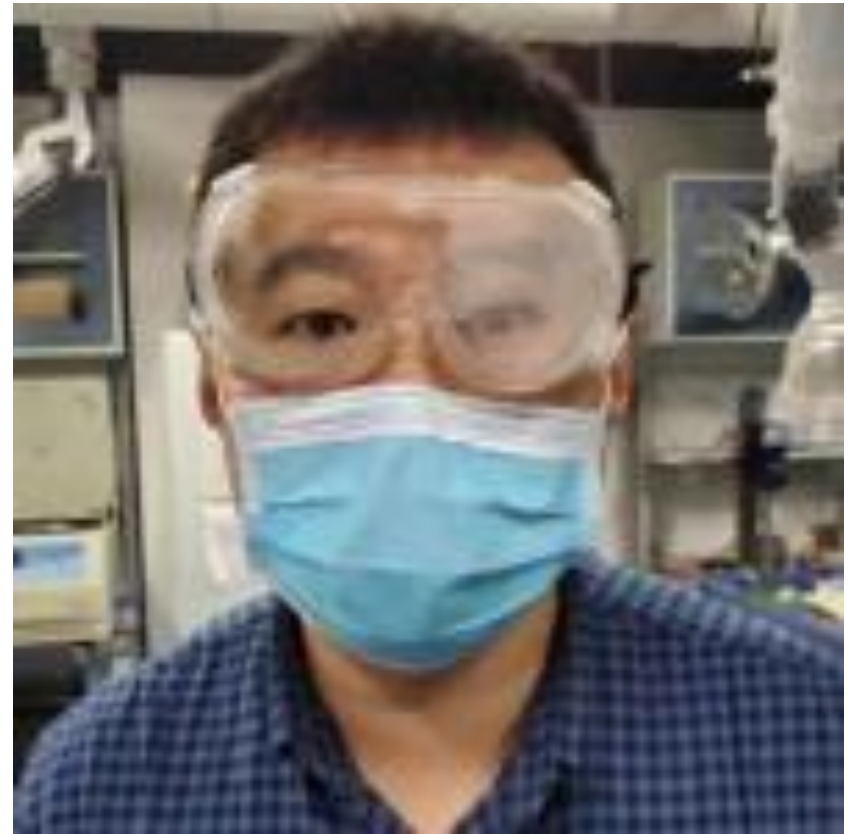
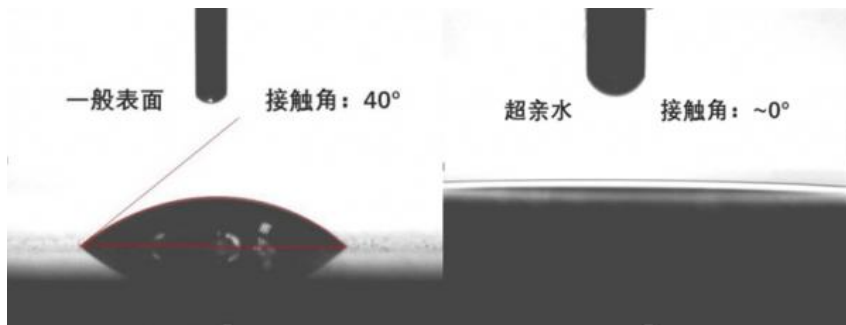
$$\text{so } 1 < (w_{ad} / \gamma_{lg}) < 2$$

If liquid doesn't wet surface

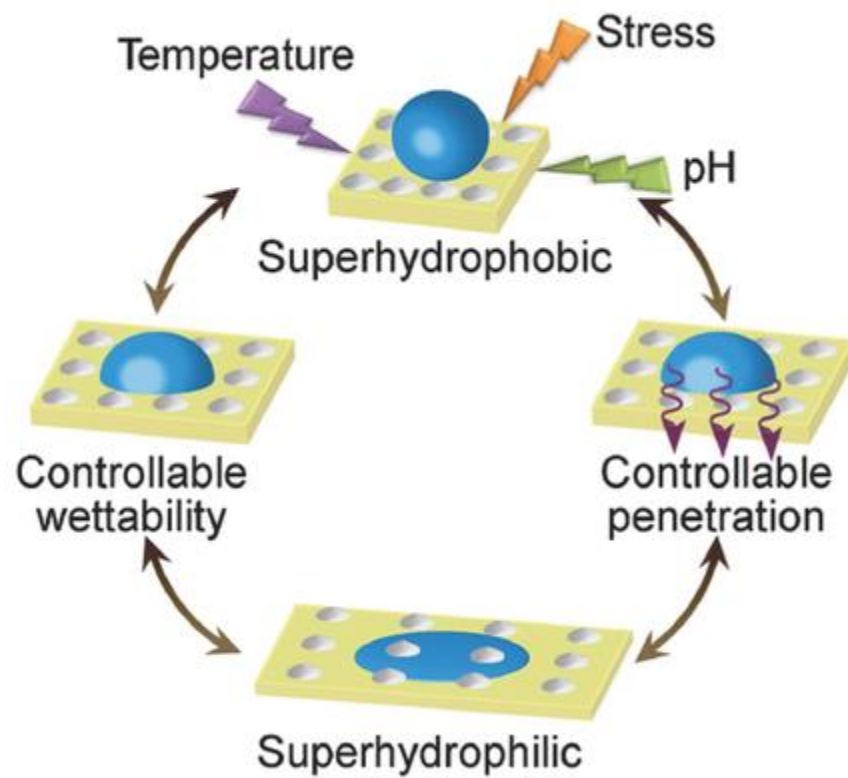
$$90^\circ < \theta_c < 180^\circ, 0 > \cos \theta_c > -1$$

$$\text{so } 0 < (w_{ad} / \gamma_{lg}) < 1$$





https://newshub.sustech.edu.cn/zh/wp-content/uploads/2020/02/2020021719562565.mp4?_=1





Summary

- Phase diagrams

$$T_{\text{trs}}: T_b, T_f, T_3, T_c$$

- The stability of phase

Phase equilibrium $\mu(\alpha) = \mu(\beta)$

Phase transition $\mu(\alpha) > \mu(\beta)$

Constant p , T increase, μ decrease

Constant T , p increase, μ increase



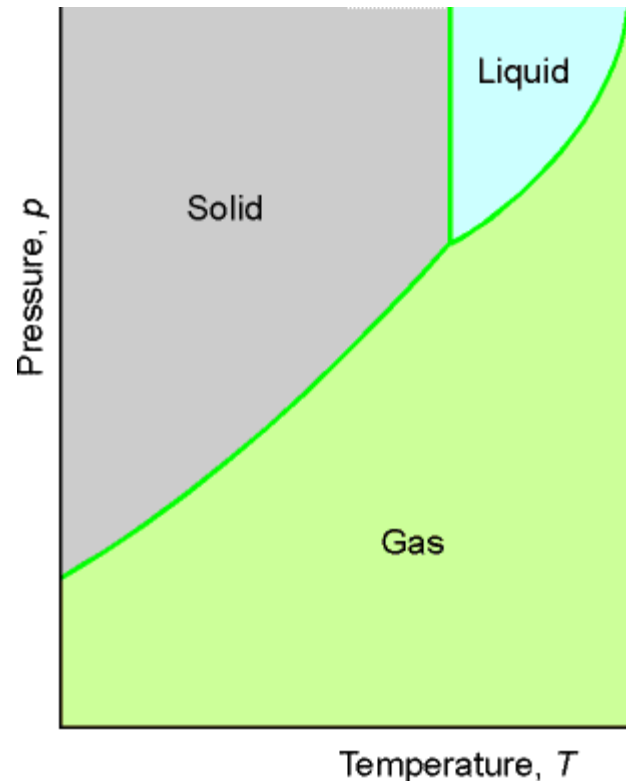
- Phase boundary

Clapeyron equation

$$\frac{dp}{dT} = \frac{\Delta_{trs}S}{\Delta_{trs}V} = \frac{\Delta_{trs}H}{T\Delta_{trs}V}$$

Liquid-vapour boundary

$$\Delta p \rightarrow \Delta T$$





- Vapour pressure

$$p = p^* e^{V_m \Delta P / RT}$$

$$\frac{p - p^*}{p^*} \approx \frac{V_m \Delta P}{RT}$$

外压变化和蒸汽压的关系

$$p = p^* e^{-\chi}$$

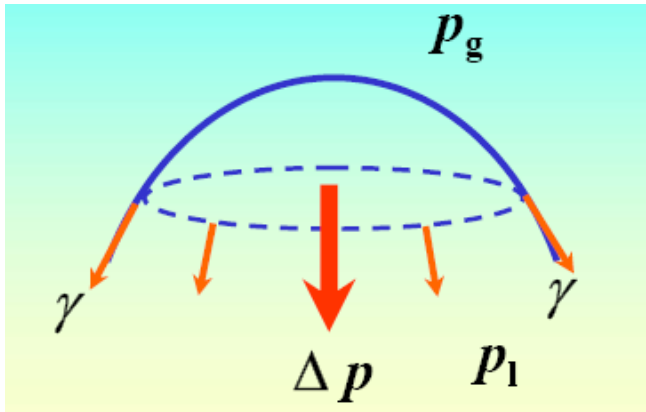
$$\chi = \frac{\Delta_{vap} H}{R} \left(\frac{1}{T} - \frac{1}{T^*} \right)$$

已知一个温度下的蒸汽压求另外一个温度下的蒸汽压（温度和蒸汽压的关系）

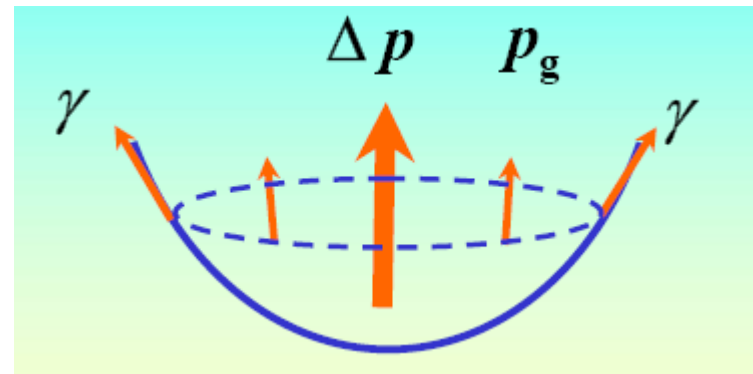


- Classification of phase transition
- Curved liquid surface

Laplace equation



$$P_l - P_g = \frac{2\gamma}{r}$$



$$P_g = P_l + \frac{2\gamma}{r}$$

$$P_l - P_g = -\frac{2\gamma}{r}$$



Klevin equation

droplet $p = p^* e^{2\gamma V_m / rRT}$

$$p > p^*$$

cavity $p = p^* e^{-2\gamma V_m / rRT}$

$$p < p^*$$

Capillary action

Contact angle

